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Electric machine having rotor with integrated fan

Abstract

Aircraft may include electric motors. The aircraft electric motors may include a rotor assembly comprises an outer diameter portion, and inner diameter portion, and an end portion, arranged to define a stator cavity, and a rotor hub configured to connect to a shaft of the aircraft electric machine, a stator assembly arranged within the stator cavity, and at least one fan assembly arranged in the rotor assembly and configured to induce a cooling flow through the stator cavity.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to electric motors, and more particularly, to electric motor assemblies with high efficiency and power density having improved cooling and structure for aircraft applications.

(2) Traditional electric motors may include a stator and a rotor, with electrical motor windings in the stator that, when energized, drive rotation of the rotor about a central axis. Permanent magnet motors are widely used for high power density and efficient applications in aviation industry. The

high torque density can be achieved by maximizing the magnetic loading through implementation of the Halbach array permanent magnet rotor structure; however, the dense permanent magnets can be a major barrier when minimizing the weight of the application. Accordingly, improved electric motor components may be used to improve the weight of such electric motors while also provide additional benefits, such as improved power density and the like.

BRIEF DESCRIPTION

(3) According to some embodiments, aircraft electric motors are provided. The aircraft electric motors include a rotor assembly comprises an outer diameter portion, and inner diameter portion, and an end portion, arranged to define a stator cavity, and a rotor hub configured to connect to a shaft of the aircraft electric machine, a stator assembly arranged within the stator cavity, and at least one fan assembly arranged in the rotor assembly and configured to induce a cooling flow through the stator cavity.

(4) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the at least one fan assembly is arranged at an outer diameter end of the end portion.

(5) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the at least one fan assembly is arranged at an inner diameter end of the end portion.

(6) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the at least one fan assembly comprises a first fan assembly arranged at an outer diameter end of the end portion and a second fan assembly arranged at an inner diameter end of the end portion.

(7) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the at least one fan assembly comprises a plurality of fan blades, wherein the plurality of fan blades provide structural support and load transfer within the rotor assembly.

(8) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that an inner diameter end of the end portion, the inner diameter portion, and the rotor hub are coupled together at a hub junction.

(9) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include a plurality of fasteners extending through and joining the end portion, the inner diameter portion, and the rotor hub together.

(10) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the outer diameter portion is coupled to the end portion by a plurality of fasteners.

(11) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the outer diameter portion comprises a set of outer diameter magnets, the inner diameter portion comprises a set of inner diameter magnets, and the end portion comprises a set of end portion magnets.

(12) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the outer diameter magnets, the inner diameter magnets, and the end portion magnets are arranged in a substantially U-shaped configuration defining the stator cavity.

(13) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the outer diameter portion comprises a flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly, wherein the flow control feature is configured to direct a cooling flow into the stator cavity.

(14) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the inner diameter portion comprises a flow

control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly, wherein the flow control feature is configured to direct a cooling flow into the stator cavity.

(15) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the outer diameter portion comprises a first flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly and the inner diameter portion comprises a second flow control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly, wherein each of the first flow control feature and the second flow control features is configured to direct a cooling flow into the stator cavity.

(16) In addition to one or more of the features described herein, or as an alternative, further embodiments of the electric motors may include that the stator assembly comprises a plurality of windings arranged within the stator cavity.

(17) According to some embodiments, aircraft are provided. The aircraft may include at least one aircraft electric machine, at least one electrical device, and a power distribution system configured to distribute power from the at least one electric machine to the at least one electrical device. The at least one aircraft electric machine may include a rotor assembly comprises an outer diameter portion, and inner diameter portion, and an end portion, arranged to define a stator cavity, and a rotor hub configured to connect to a shaft of the aircraft electric machine, a stator assembly arranged within the stator cavity, and at least one fan assembly arranged in the rotor assembly and configured to induce a cooling flow through the stator cavity.

(18) In addition to one or more of the features described herein, or as an alternative, further embodiments of the aircraft may include that the at least one fan assembly comprises at least one of a fan assembly arranged at an outer diameter end of the end portion and a fan assembly arranged at an inner diameter end of the end portion.

(19) In addition to one or more of the features described herein, or as an alternative, further embodiments of the aircraft may include that the at least one fan assembly comprises a plurality of fan blades, wherein the plurality of fan blades provide structural support and load transfer within the rotor assembly.

(20) In addition to one or more of the features described herein, or as an alternative, further embodiments of the aircraft may include that the outer diameter portion comprises a set of outer diameter magnets, the inner diameter portion comprises a set of inner diameter magnets, and the end portion comprises a set of end portion magnets.

(21) In addition to one or more of the features described herein, or as an alternative, further embodiments of the aircraft may include that the outer diameter portion comprises a first flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly and configured to direct a cooling flow into the stator cavity along an outer diameter surface of the stator assembly, and the inner diameter portion comprises a second flow control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly and configured to direct a cooling flow into the stator cavity along an inner diameter surface of the stator assembly.

(22) In addition to one or more of the features described herein, or as an alternative, further embodiments of the aircraft may include that the stator assembly comprises a plurality of windings arranged within the stator cavity.

(23) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. Features which are described in the context of separate aspects and embodiments may be used together and/or be interchangeable. Similarly, features described in the context of a single embodiment may also be provided separately or in any suitable subcombination. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be

understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various features will become apparent to those skilled in the art from the following detailed description of the disclosed non-limiting embodiments. The drawings that accompany the detailed description can be briefly described as follows:
- (2) FIG. 1A is a partial view of an embodiment of electric motor;
- (3) FIG. 1B is a cross-sectional view of an embodiment of a stator core of the electric motor of FIG. 1A;
- (4) FIG. 2A is a schematic illustration of an aircraft electric motor in accordance with an embodiment of the present disclosure;
- (5) FIG. 2B is a side elevation view of the aircraft electric motor of FIG. 2A;
- (6) FIG. 2C is a partial cut-away illustration of the aircraft electric motor of FIG. 2A;
- (7) FIG. 2D is a separated-component illustration of the aircraft electric motor of FIG. 2A;
- (8) FIG. 3A is a schematic illustration of a rotor and stator of an aircraft electric motor in accordance with an embodiment of the present disclosure;
- (9) FIG. 3B is a schematic illustration of the rotor and stator of FIG. 3A as arranged within a rotor sleeve in accordance with an embodiment of the present disclosure;
- (10) FIG. 4A is a schematic illustration of a radial cross-section of an aircraft electric motor in accordance with an embodiment of the present disclosure;
- (11) FIG. 4B is an enlarged illustration of a portion of the structure shown in FIG. 4A;
- (12) FIG. 5A is a schematic illustration of a rotor assembly in accordance with an embodiment of the present disclosure;
- (13) FIG. 5B is a schematic illustration of a portion of the rotor assembly of FIG. 5A;
- (14) FIG. 5C is an enlarged sectional illustration of the portion of the rotor assembly shown in FIG. 5B;
- (15) FIG. 6 is a schematic illustration of a portion of an electric machine in accordance with an embodiment of the present disclosure;
- (16) FIG. 7A is a schematic illustration of a portion of an electric machine in accordance with an embodiment of the present disclosure;
- (17) FIG. 7B is a schematic illustration of cooling flow passing through the portion of the electric machine shown in FIG. 7A;
- (18) FIG. 8A is a schematic illustration of a portion of a rotor assembly in accordance with an embodiment of the present disclosure having a fan assembly thereon;
- (19) FIG. 8B is a schematic illustration of a fan blade of the fan assembly of FIG. 8A; and
- (20) FIG. 9 is a schematic view of a power system of an aircraft that may employ embodiments of the present disclosure.
- (21) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, that the following description and drawings are intended to be illustrative and explanatory in nature and non-limiting.

DETAILED DESCRIPTION

- (22) Referring to FIGS. 1A-1B, schematic illustrations of an electric motor **100** that may incorporate embodiments of the present disclosure are shown. FIG. 1A illustrates a cross-sectional view of the electric motor **100** and FIG. 1B illustrates a cross-sectional view of a stator core of the

electric motor **100**. The electric motor **100** includes a rotor **102** configured to rotate about a rotation axis **104**. A stator **106** is located radially outboard of the rotor **102** relative to the rotation axis **104**, with a radial airgap **108** located between the rotor **102** and the stator **106**. As illustrated, the rotor **102** may be mounted on a shaft **110** which may impart rotational movement to the rotor **102** or may be driven by rotation of the rotor **102**, as will be appreciated by those of skill in the art. The rotor **102** and the shaft **110** may be fixed together such that the rotor **102** and the shaft **110** rotate about the rotation axis **104** together as one piece.

(23) The stator **106** includes a stator core **112** in which a plurality of electrically conductive stator windings **114** are disposed. In some embodiments, such as shown in FIG. **1A**, the stator core **112** is formed from a plurality of axially stacked laminations **116**, which are stacked along the rotation axis **104**. In some embodiments, the laminations **116** are formed from a steel material, but one skilled in the art will readily appreciate that other materials may be utilized. The stator windings **114**, as shown, include core segments **118** extending through the stator core **112** and end turn segments **120** extending from each axial stator end **122** of the stator core **112** and connecting circumferentially adjacent core segments **118**. When the stator windings **114** are energized via an electrical current therethrough, the resulting field drives rotation of the rotor **102** about the rotation axis **104**. Although FIG. **1A** illustrates the stator core **112** arranged radially inward from the stator windings **114**, it will be appreciated that other configurations are possible without departing from the scope of the present disclosure. For example, in some embodiments, the stator structure may be arranged radially inward from a rotating rotor structure.

(24) FIG. **1B** is an axial cross-sectional view of the stator core **112**. Each lamination **116** of the stator core **112** includes a radially outer rim **124** with a plurality of stator teeth **126** extending radially inwardly from the outer rim **124** toward the rotation axis **104**. Each of the stator teeth **126** terminate at a tooth tip **128**, which, together with a rotor outer surface **130** (shown in FIG. **1A**) of the rotor **102**, may define the radial airgap **108**. Circumferentially adjacent stator teeth **126** define an axially-extending tooth gap **132** therebetween. Further, in some embodiments, a plurality of stator fins **134** extend radially outwardly from the outer rim **124**.

(25) Electric motors, as shown in FIGS. **1A-1B** may require cooling due to high density configurations, various operational parameters, or for other reasons. For example, high-power-density aviation-class electric motors and drives may require advanced cooling technologies to ensure proper operation of the motors/drives. These machines are generally thermally limited at high power ratings and their performance can be improved by mitigating thermal limitations. To maintain desired temperatures, a thermal management system (TMS) is integrated into the system, which provides cooling to components of the system. Onboard an aircraft, power requirements, and thus thermal management system (TMS) loads, are substantially higher during takeoff. Sizing of the TMS for takeoff conditions (i.e., maximum loads) results in a TMS having a high weight to accommodate such loads. This results in greater weight and lower power density during cruise conditions which do not generate such loads, and thus does not require a high cooling capacity TMS. Balancing weight constraints and thermal load capacities is important for such aviation applications.

(26) In view of such considerations, improved aviation electric motors are provided herein. The aviation electric motors or aircraft electric motors, described herein, incorporate lightweight materials and compact design to reduce weight, improve thermal efficiencies, improve power efficiencies, and improve power density.

(27) Turning now to FIGS. **2A-2D**, schematic illustrations of an aircraft electric motor **200** in accordance with an embodiment of the present disclosure are shown. FIG. **2A** is an isometric illustration of the aircraft electric motor **200**, FIG. **2B** is a side elevation view of the aircraft electric motor **200**, FIG. **2C** is a partial cut-away view illustrating internal components of the aircraft electric motor **200**, and FIG. **2D** is a schematic illustration of components of the aircraft electric motor **200** as separated from each other. The aircraft electric motor **200** includes a motor housing

202, a cooling system **204**, a first power module system **206**, and a second power module system **208**.

(28) The motor housing **202** houses a stator **210** and a rotor **212**, with the rotor **212** configured to be rotatable about the stator **210**. In this illustrative embodiment, the rotor **212** includes a U-shaped magnet **214** arranged within a similarly shaped U-shaped rotor sleeve **216**. The rotor sleeve **216** is operably connected to a hub **218**. The hub **218** is fixedly attached to a first shaft **220**. The first shaft **220** is operably connected to a second shaft **222**. In some configurations, the first shaft **220** may be a high speed shaft and may be referred to as an input shaft. In such configurations, the second shaft **222** may be a low speed shaft and may be referred to as an output shaft. The connection between the first shaft **220** and the second shaft **222** may be by a gear assembly **224**, as described herein.

(29) The cooling system **204** is configured to provide cooling to the components of the aircraft electric motor **200**. The cooling system **204**, as shown in FIG. 2D, includes a heat exchanger **226** and a header **228**. The heat exchanger **226** and the header **228** may form a closed-loop cooling system that may provide air-cooling to a working fluid at the heat exchanger **226**. The header **228** may be, in some configurations, a two-phase di-electric cooling header. A cooled working fluid may be pumped from the heat exchanger **226** into the header **228** using a pump **229** and distributed into embedded cooling channels **230** that are arranged within the stator **210**. As the aircraft electric motor **200** is operated, heat is generated and picked up by the working fluid within the embedded cooling channels **230**. This heated working fluid is then passed through the header **228** back to the heat exchanger **226** to be cooled, such as by air cooling. Although described as air-cooling, other cooling processes may be employed without departing from the scope of the present disclosure.

(30) As shown, the heat exchanger **226** of the cooling system **204** may be a circular or annular structure that is arranged about the motor housing **202**. This configuration and arrangement allows for improved compactness of the system, which may be advantageous for aircraft applications. The rotor sleeve **216** with the magnets **214**, the stator **210**, and the gear assembly **224** fit together (although moveable relative to each other) within the motor housing **202**, providing for a compact (low volume/size) design.

(31) As noted above, the rotor sleeve **216** may be operably coupled to a first shaft **220** by the hub **218**. The first shaft **220** may be operably coupled to a first gear element **232** and the second shaft **222** may be operably coupled to a second gear element **234**. The first and second gear elements **232**, **234** may form the gear assembly **224**. The first and second gear elements **232**, **234** are arranged to transfer rotational movement from the first shaft **220**, which is driven in rotation by the hub **218** and the rotor sleeve **216** of the rotor **212**, to the second shaft **222**. In some embodiments, the first shaft **220** may be operably connected to a sun gear as the first gear element **232** that engages with a plurality of planetary gears and drives rotation of the second gear element **234** which may be operably connected to the second shaft **222**. In some embodiments, the second shaft **222** may be connected to a fan or other component to be rotated by the aircraft electric motor **200**.

(32) The aircraft electric motor **200** includes the first power module system **206** and the second power module system **208**. The first and second power module systems **206**, **208** can include capacitors and other electronics, including, but not limited to, printed circuit boards (PCBs) that are configured to control and operate the aircraft electric motor **200**. Again, the profile of the aircraft electric motor **200** of the present disclosure presents a low profile or compact arrangement that reduces the volume of the entire power system, which in turn can provide for improved weight reductions. In some embodiments, the first and second power module systems **206**, **208** may be electrically connected to the stator **210** to cause an electric current therein. As the electric current will induce an electromagnetic field which will cause the rotor **212** to rotate.

(33) Referring now to FIGS. 3A-3B, schematic illustrations of a portion of an aircraft electric motor **300** in accordance with an embodiment of the present disclosure is shown. FIGS. 3A-3B illustrate a portion of a rotor **302** and a stator **304** of the aircraft electric motor **300**. FIG. 3A illustrates the rotor **302** and the stator **304** and FIG. 3B illustrates these components arranged

within a rotor sleeve **306**.

(34) The rotor **302** is formed of a plurality of U-shaped magnets **308**. In some configurations, the plurality of magnets **308** can be arranged with alternating polarity in a circular structure. Arranged within the “U” of the U-shaped magnets **308** is the stator **304**. The stator **304** is formed of a plurality of windings **310**. In this configuration, the windings **310** are arranged with a header **312**. The header **312** may be part of a cooling system, such as that shown and described above. The header **312** can be configured to cycle a working fluid through cooling channels **314** for cooling of the windings **310**, as shown in FIG. 3B.

(35) The windings **310** may be wrapped about a support structure **316** (e.g., back iron or yoke). The support structure **316**, in some embodiments and as shown in FIG. 3B, may include a laminate portion **318** and a magnetic portion **320**. In some such embodiments, the laminate portion **318** may be formed from cobalt steel laminate and the magnetic portion **320** may be formed from a soft magnetic composite. The laminate portion **318** may be provided to capture in-plane flux from outer and inner rotor. The magnetic portion **320** may be provided to capture end rotor flux and may take a shape/filler in a gap through the end turns of the coil. The windings **310** include end connections **322** and may be electrically connected to one or more power module systems of the aircraft electric motor, such as shown above.

(36) As shown in FIG. 3B, the magnets **308** are U-shaped and arranged within the rotor sleeve **306**. The rotor sleeve **306** is a substantially U-shaped sleeve that is sized and shaped to receive the U-shaped magnets **308**. In this illustrative configuration, the rotor sleeve **306** can include an inner sleeve **324**. The inner sleeve **324** may be configured to provide support to a portion of the magnets **308**. It will be appreciated that there is no direct contact between the windings **310** and the magnets **308**. This lack of contact allows for free rotation of the rotor **302** relative to the stator **304** during operation.

(37) In aviation-class electric motors, such as shown and described above, a high-power density can be achieved by maximizing torque at a given speed. The torque density can be increased by improving utilization of magnetic materials and increase magnetic loading. Prior concepts for maximizing power density was achieved through minimizing the core of the rotor system. However, such minimization has an impact on magnetic loading (average airgap flux density). Conventionally, introducing a magnetic tooth can increase magnetic loading but may also increase torque ripple. Torque ripple is an effect seen in electric motor designs and refers to a periodic increase or decrease in output torque as the motor shaft rotates. Accordingly, it is desirable to both maximize magnetic loading while minimizing torque ripple. In view of this, embodiments of the present disclosure are directed to incorporating non-magnetic teeth and/or non-magnetic back iron, yoke, or support structure within the motor assembly. The non-magnetic structures (teeth and/or support structure) are made from non-magnetic materials (e.g., potting material, ceramic, etc.) may be infused or embedded with magnetic wires. In accordance with embodiments of the present disclosure, the introduction of magnetic wire-infused teeth and/or support structures results in reduced weight and improved power density. Further, advantageously, such configurations can provide a low weight solution without sacrificing average torque of the motor. Shaping of the wires near an airgap (e.g., to the magnets of the motor) can also help manipulate the harmonics in the airgap and result in redistribution of torque ripple harmonics and reduce torque ripple without impacting average torque.

(38) Referring to FIGS. 4A-4B, schematic illustrations of a portion of an aircraft electric motor **400** in accordance with an embodiment of the present disclosure is shown. FIGS. 4A-4B illustrate a portion of a rotor **402** and a stator **404** of the aircraft electric motor **400**. FIG. 4A illustrates the full circular structure of the rotor **402** and the stator **404** and FIG. 4B illustrates an enlarged illustration of a portion of the rotor **402** and the stator **404**. The rotor **402** and the stator **404** may be part of an aircraft electric motor similar to that shown and described herein and used as described herein.

(39) As shown, the rotor **402** is arranged about the stator **404**, with an outer portion **402a** and an

inner portion **402b** arranged radially outward and inward from the stator **404**, respectively. The outer and inner portions **402a**, **402b** may be parts of a substantially U-shaped magnet assembly, as shown and described above. The stator **404** is arranged between the outer and inner portions **402a**, **402b** with an airgap **406** therebetween, as shown in FIG. **4B**. The rotor **402** includes a plurality of magnets **408**, which may be substantially U-shaped and span from the outer portion **402a** to the inner portion **402b**. An outer rotor sleeve **410** and an inner rotor sleeve **412** may be separate components or a continuous structure, as shown and described above, and are configured to support and retain the magnets **408** of the rotor **402**. Further, one or more retention sleeves **414** may be arranged on a side of the magnets **408** that faces the stator **404**. The rotor **402** is configured to be rotationally driven by current that is passed through the stator **404**.

(40) In some configurations, the stator **404** may include a support structure **416** (e.g., a back iron or yoke). The support structure **416** may be configured to support, on a radial outer side thereof, a plurality of outer teeth **418**, outer coils **420**, and outer cooling channels **422**. Similarly, on a radially inner side of the support structure **416** are arranged a plurality of inner teeth **424**, inner coils **426**, and inner cooling channels **428**. It will be appreciated that, in accordance with some embodiments, the support structure **416** may be omitted and the structural aspects thereof provided by other features, as shown and described herein.

(41) In some embodiments of the present disclosure, one or more of the outer teeth **418**, the inner teeth **424**, and/or the support structure **416** may be made of a non-magnetic material with embedded magnetic wires. In some example embodiments, each of the outer teeth **418** the inner teeth **424**, and/or the support structure **416** may be formed of a non-magnetic material with embedded magnetic wires and shaped to reduce torque ripple while increasing magnetic loading and improving manufacturability and address stack-up tolerance challenges.

(42) As shown in FIG. **4A**, the rotor **402** and stator **404** form a substantially ring-shape or annular shape. As shown, the outer teeth **418** and the inner teeth **424** are each arranged in a circumferential arrangement and extend radially from the support structure **416**. The outer teeth **418** extend radially outward from the support structure **416** and the inner teeth **424** extend radially inward from the support structure **416**. In some configurations, the teeth **418**, **424** may be the same in shape, orientation, material, and the like about the circumferences of the stator **404**. In other embodiments, the teeth **418**, **424** may be arranged in sets or specific configurations arranged in a repeating pattern about the respective circumferential arrangement.

(43) In aviation-class electric machines, high power density electric power trains can be achieved by fully integrating electrical, thermal, and mechanical functionality to take advantage of various phenomenon. While maximizing power density may be desirable, such increases in power density may inherently increase losses and reduce efficiency. Therefore, it is critical to extract heat loads in an efficient manner with minimal impact on the overall performance.

(44) To increase power density, the introduction of various elements such as use of magnetic teeth and an inner sleeve for the outer rotor have been implemented. However, such increased power density features may result in space harmonics in the airgap that can impact rotor operation and/or introduce Eddy current losses in the sleeve of the rotor. These losses may result in an increase in rotor temperature. The increased rotor temperature may impact permanent magnets that are housed in the rotor. In view of this and other considerations, embodiments of the present disclosure are directed to integration of a fan into the rotor structure.

(45) For example, in some embodiments a fan, fan structure, or fan assembly (hereinafter referred to as a fan assembly) may be arranged between portions of the rotor (e.g., outer rotor portion and end rotor portion). In other embodiments, such a fan assembly may be arranged between the end rotor portion and an inner rotor portion. In other embodiments, multiple fan assemblies may be arranged at different locations on the rotor. Improved airflow provided by the fan assemblies may enhance extraction of heat loads generated due to electromagnetic losses from the rotor and stator sub-components. In some embodiments, the fan assemblies may be configured to provide structural

support and loads transfers between elements of the rotor.

(46) Referring now to FIGS. 5A-5C, schematic illustrations of a rotor **500** for an electric machine in accordance with an embodiment of the present disclosure are shown. The rotor **500** may be configured to be used with an electric machine or electric motor, such as the systems shown and described above. In the above illustrated embodiments, the rotors are substantially unitary in form with U-shaped magnets installed therein. In contrast, the rotor **500** has a multi-part construction, although the magnet assembly substantially corresponds to a U-shaped magnet assembly, and thus the functionality of the rotor **500** may be substantially similar to that shown and described above.

(47) The rotor **500** includes an outer diameter portion **502**, an inner diameter portion **504**, and an end portion **506**. As shown the outer diameter portion **502** and the inner diameter portion **504** are oriented in an axial direction and the end portion **506** is oriented in a radial direction. Each of the outer diameter portion **502** and the inner diameter portion **504** may be fixedly coupled to the end portion **506** by sets of fasteners **508**. A rotor hub **510** may extend radially inward from the inner diameter portion **504**, and may be coupled between the inner diameter portion **504** and the end portion **506** and fastened thereto by the fasteners **508** at a hub junction **511**. The outer diameter portion **502** includes outer diameter magnets **512**, the inner diameter portion includes inner diameter magnets **514**, and the end portion **506** includes end portion magnets **516**. The arrangement of the magnets **512**, **514**, **516** may substantially match a U-shaped magnet configuration, such as shown and described above. In some configurations, the end portion magnets **516** may be omitted, and the rotor **500** may only include the outer diameter magnets **512** in the outer diameter portion **502** and the inner diameter magnets **514** of the inner diameter portion.

(48) The outer diameter portion **502** of the rotor **500** includes a rotor wrap **518** that radially constrains the outer diameter portion **502** during operation. The outer diameter magnets **512** are arranged radially inward from the rotor wrap **518** and are secured within an outer diameter support structure **520** that may retain the outer diameter magnets **512** in a radial direction (both radially inward and radially outward), with the outer diameter support structure **520** having axial portions on axial ends of the set of outer diameter magnets **512**. The fasteners **508** may engage with an axial portion of the outer diameter support structure **520** to secure the outer diameter portion **502** to the end portion **506** of the rotor **500**. The outer diameter support structure **520** may provide a full wrap structure about the outer diameter magnets **512**, with the rotor wrap **518** providing additional support to prevent radial movement of the outer diameter portion **502** during operation.

(49) The inner diameter portion **504** of the rotor **500** houses the inner diameter magnets **514** within an inner diameter support structure **522**. The inner diameter support structure **522** is configured to retain the inner diameter magnets **512** in a radial direction (both radially inward and radially outward) and in the axial direction. The fasteners **508** may engage with an axial portion of the inner diameter support structure **522** to secure the inner diameter portion **504** to the end portion **506** of the rotor **500**.

(50) The end portion **506** of the rotor **500** includes an end portion support structure **524** that houses the end portion magnets **516**. The extending radially outward from the end portion support structure **524** is an outer diameter extension **526** and extending radially inward from the end portion support structure **524** is an inner diameter extension **528**. Each of the outer diameter extension **526** and the inner diameter extension **528** include apertures for receiving the fasteners **508** for engagement with the outer diameter portion **502** and the inner diameter portion **504**. In this illustrative configuration, the end portion **506** includes an outer diameter fan assembly **530** arranged between the end portion support structure **524** and the outer diameter extension **526** and an inner diameter fan assembly **532** arranged between the end portion support structure **524** and the inner diameter extension **528**.

(51) When assembled together, the outer diameter portion **502**, the inner diameter portion **504**, and the end portion **506** of the rotor **500** define a stator cavity **534** for receiving a stator structure, such

as the windings, cooling elements, and the like, such as shown and described above (e.g., stator **304** shown in FIGS. **3A-3B**). FIG. **5B** illustrates the end portion **506** of the rotor **500** in isolation. FIG. **5C** illustrates an enlarged, sectional view of the end portion **506**. As shown in FIG. **5C**, each of the outer diameter fan assembly **530** and the inner diameter fan assembly **532** include a plurality of fan blades **536**. The fan blades **536** of the fan assemblies **530**, **532** are arranged with flow paths **538** defined between adjacent fan blades **536** to permit airflow to pass through the end portion **506**.

(52) By including the flow paths **538** in the end portion **506** of the rotor **500**, airflow may be able to pass through the stator cavity **534** and provide cooling to the components of the stator and to the magnets **512**, **514**, **516** of the rotor **506**. That is, a cooling flow may be directed into the stator cavity **534** at a forward end (e.g., end in which the stator is installed into the stator cavity **534**) and the cooling flow may exit through the flow paths **538**. This additional cooling may provide additional operational benefits to the electric machine in which the rotor **500** is incorporated. For example, the throughflow through the fan assemblies **530**, **532** may ensure a level pressure drop to reduce uneven flow through and around the stator arranged within the stator cavity **534**. Further, the throughflow may extract heat from both surfaces of the rotor **500** and the stator components arranged within the stator cavity **534** and may minimize hot spots on both structures. Further, the throughflow through the fan assemblies **530**, **532** may result in reduced magnet operating temperatures and can enable an increased power density of the electric machine.

(53) It will be appreciated that the direction of rotation and/or the angle/orientation of the fan blades of the fan assemblies **530**, **532** may control a flow direction of cooling air into and through the stator cavity **534**. For example, in some configurations, the direction of flow of the cooling air, that is induced by the fan assemblies **530**, **532**, may be in a direction from the open end of the stator cavity **534** toward the end portion **506**. If the direction of rotation (with the same oriented fan blades) is changed to be opposite of such a configuration, or the fan blades are reversed in orientation, the flow direction may be from the end portion **506** toward the open end of the stator cavity **534**. As such, the direction of flow induced by the fan assemblies of the present disclosure is not intended to be limited to only one directional configuration.

(54) Referring still to FIGS. **5A-5C**, in accordance with a non-limiting example, the rotor wrap **518** may be formed from carbon fiber or other non-magnetic material. In some embodiments, the magnets **512**, **514**, **516** may be Neodymium magnets or Samarium-cobalt magnets. In a non-limiting configuration, the outer diameter magnets and the end portion magnets may be Neodymium magnets and the inner diameter magnets may be Samarium-cobalt magnets. In some embodiments, the structure of the outer diameter portion **502** that supports the outer diameter magnets **512** may be formed from a non-magnetic metal, such as aluminum. In some embodiments, the structure of the inner diameter portion **504** that supports the inner diameter magnets **514** may be formed from titanium alloys, Inconel alloys, or the like. In some embodiments, the structure of the end portion **506** that supports the end portion magnets **516** may be formed from titanium alloys, Inconel alloys, or the like. In some embodiments, the end portion **506** may include a similar warp (e.g., similar to rotor wrap **518**) to provide radial constraint upon the end portion **506**. Although certain example materials are described for forming various features and structures of the electric machine components of the present disclosure, such materials are not intended to be limiting but rather are provided as examples of material that may be employed in manufacturing the described components.

(55) Referring now to FIG. **6**, a schematic illustration of a portion of an electric machine **600** in accordance with an embodiment of the present disclosure is shown. The electric machine **600** includes a stator assembly **602** and a rotor assembly **604**. The stator assembly **602** may be configured similar to that shown and described above, having windings, cooling channels, and the like. A portion of the stator assembly **602** is arranged to fit within a stator cavity of the rotor assembly **604** (e.g., stator cavity **534** shown in FIG. **5A**).

(56) The rotor assembly **604** includes a rotor hub **606** that may connect to a shaft or the like that

may be rotationally driven by the rotor assembly **604**. The rotor assembly **604**, similar to that shown in FIGS. 5A-5C, includes an outer diameter portion **608**, an inner diameter portion **610**, and an end portion **612**. The rotor hub **606**, the inner diameter portion **610**, and the end portion **612** may be joined together at a hub junction **614**. One or more fasteners may be used to join the rotor hub **606**, the inner diameter portion **610**, and the end portion **612** together. In other configurations, adhesives, welding, bonding, or mechanical connections (e.g., tongue-and-groove, latches, etc.) may be used to join the components together. Further still, in some embodiments, two or more of the rotor hub **606**, the inner diameter portion **610**, and the end portion **612** may be integrally formed (e.g., cast, molded, etc.) as a unitary structure. At the outer diameter of the rotor assembly **604**, the outer diameter portion **608** is coupled to the end portion **612** in a similar fashion (e.g., fasteners, mechanical connections, unitary formation, etc.).

(57) The outer diameter portion **608** includes outer diameter magnets **616**. The inner diameter portion **610** includes inner diameter magnets **618**. The end portion **612** includes end portion magnets **620**. Similar to the configuration of FIGS. 5A-5C, the magnets **616**, **618**, **620** are arranged to form a substantially U-shaped configuration, with the stator assembly **602** arranged within the open portion of the U-shaped structure. In this configuration, the rotor assembly **604** includes one fan assembly **622**. The fan assembly **622** is arranged at the outer diameter end of the end portion **612**. That is, in contrast to the configuration of FIGS. 5A-5C which includes inner and outer diameter fan assemblies, the rotor assembly **604** of FIG. 6 includes only the single fan assembly **622** which is arranged at the outer diameter of the end portion **612** of the rotor assembly **604**.

(58) Referring now to FIGS. 7A-7B, schematic illustrations of a portion of an electric machine **700** in accordance with an embodiment of the present disclosure are shown. FIG. 7A illustrates a side, sectional view of the portion the electric machine **700** and FIG. 7B is an enlarged illustration showing cooling flow through the portion of the electric machine **700**. The electric machine **700** includes a stator assembly **702** and a rotor assembly **704**. The stator assembly **702** may be configured similar to that shown and described above, having windings, cooling channels, and the like. A portion of the stator assembly **702** is arranged to fit within a stator cavity **706** of the rotor assembly **704**.

(59) The rotor assembly **704** includes a rotor hub **708** that connected to a shaft **710** that may be rotationally driven by the rotor assembly **704**. The rotor assembly **704** includes an outer diameter portion **712** and an inner diameter portion **714**. An end portion **716** connects the rotor hub **708**, the inner diameter portion **714**, and the outer diameter **712** together, as shown and described above. Each of the outer diameter portion **712**, the inner diameter portion **714**, and the end portion **716** include respective magnets, as shown and described above. The outer diameter portion **712**, the inner diameter portion **714**, and the end portion **716** and the magnets thereof are arranged to form a substantially U-shaped configuration, with the stator assembly **702** arranged within the open portion of the U-shaped structure which defines stator cavity **706**. In this configuration, the rotor assembly **704** includes a fan assembly **718** defined in the end portion **716** and arranged toward the outer diameter end of the end portion **716**.

(60) In this configuration, the rotor assembly **704** includes one or more optional flow control features to aid in directing cooling flow into and through the stator cavity **706**. For example, in this illustrative configuration, the outer diameter portion **712** includes a first flow control feature **720** and the inner diameter portion **714** includes a second flow control feature **722**. The first flow control feature **720** extends axially outward from an end of the outer diameter portion **712** opposite the end portion **716**. The second flow control feature **722** extends axially outward from an end of the inner diameter portion **714** opposite the end portion **716**. The flow control features **720**, **722** may be configured to draw in and direct flow into the stator cavity **706** to increase a flow of cooling air along surfaces of the stator assembly **702** and surfaces of the rotor assembly **704**. The cooling air directed into the stator cavity **706** may be drawn through the stator cavity **706** by the fan assembly **718**. For example, during operation, the fan assembly **718** will be rotated with the rotor

assembly **704** and the fan blades will thus be rotated to cause a pressure differential and draw the cooling flow through the stator cavity **706**.

(61) Referring to FIG. 7B, an enlarged illustration of a cooling flow **724** through the electric machine **700** is shown. The cooling flow **724** will enter the stator cavity **706** at an axial forward end, flow through the stator cavity **706** around and along surfaces of the stator assembly **702** and the rotor assembly **704**. The cooling flow **724** will then exit through the fan assembly **718** as the cooling flow **724** is pulled through the stator cavity **706** by the rotating fan blades of the fan assembly **718**. At the axial forward end of the rotor assembly **704**, the first control feature **720** will aid in directing the cooling flow **724** into the stator cavity **706** at an outer diameter portion of the stator cavity **706**. Similarly, the second control feature **722** will aid in directing the cooling flow **724** into the stator cavity **706** at an inner diameter portion of the stator cavity **706**. The directed cooling flow **724** will then pass into and through the stator cavity **706** and exit through the fan assembly **718**. In this configuration, because only one fan assembly **718** is provided, the cooling flow **724** that enters at the inner diameter side of the stator cavity **706** will be turned and flow radially outward (arrow **726**) along a surface of the end portion **706** and provide cooling to the end portion magnets and an axially aft end of the stator assembly **702** that is arranged within the stator cavity **706**. The axially turned flow will then be pulled axially aftward by the rotating fan assembly **718** and directed out of the stator cavity **706**.

(62) In the illustrative configuration of FIGS. 7A-7B, the first flow control feature **720** is configured with a curved or contoured surface **728** and the second flow control feature **722** is configured with a straight or angled surface **730**. In both configurations, the surfaces **728**, **730** are angled or contoured to funnel and direct the cooling flow **724** into the stator cavity **706**. It will be appreciated that any type of surface or geometry may be used for the flow control features **720**, **722**, and in some embodiments, but flow control features may have curved/contoured or straight/angled surfaces. Further, in some embodiments, only one of the two flow control features **720**, **722** may be employed (e.g., only the outer diameter flow control feature **720** or only the inner diameter flow control feature **722**). The flow control features **720**, **722** may be separate components that are affixed or otherwise attached to the respective portions **712**, **714** of the rotor assembly **704**. Such attachment may be by fastener, welding, bonding, adhesives, clips, clasps, tongue-and-groove connections, or the like. In other embodiments, the flow control features **720**, **722** may be integrally formed as part of the structure of the respective portions **712**, **714** of the rotor assembly **704**.

(63) As shown in FIGS. 6, 7A-7B, a rotor assembly having a configuration with a single fan assembly (outer diameter location) are illustrated. In FIGS. 5A-5C, a rotor assembly having two fan assemblies (both inner and outer diameter locations) is illustrated. In still further embodiments of the present disclosure, a rotor assembly may be provided with a single fan assembly arranged at an inner diameter end of the end portion (e.g., opposite the configuration of FIGS. 6, 7A-7B).

(64) Referring now to FIGS. 8A-8B, schematic illustrations of a portion of a rotor assembly **800** in accordance with an embodiment of the present disclosure are shown. FIG. 8A illustrates a portion of an end portion **802** of the rotor assembly **800** which may be arranged with inner and outer diameter portions to form the rotor assembly **800**, as shown and described above. FIG. 8B illustrates an enlarged illustration of a fan blade **804** of the end portion **802**. As shown in FIG. 8A, a plurality of the fan blades **804** extend radially outward from the end portion **802**. The fan blades **804** are defined by a leading edge **806**, a trailing edge **808**, a pressure side **810**, and a suction side **812**. The leading edge **806** is oriented in the direction of rotation such that the leading edge **806** of the fan blades **804** will cut into cooling flow (e.g., as shown in FIG. 7B) and draw the cooling flow along the surfaces of the fan blades **804**. The orientation and angle of the fan blades **804** may be set to ensure a desired throughflow of cooling air that is pulled through a stator cavity and directed axially downstream from the rotor assembly.

(65) Furthermore, as discussed above with respect to FIGS. 5A-5C, the direction of flow is not limited to a single direction. For example, in FIG. 8A, the flow direction is to the right on the page

(e.g., axially to the right). However, by reversing the direction of rotation of the end portion **802** or reversing the orientation/direction of the fan blades **804** with respect to the end portion **802**, the flow direction may be the opposite of that shown. That is, the fan assemblies of the present disclosure are configured to induce a cooling flow through a stator cavity, but the direction of flow is not intended to be limited to the illustrative, example configurations.

(66) Referring now to FIG. **9**, a power system **900** of an aircraft **902** is schematically shown. The power system **900** includes one or more engines **904**, one or more electric machines **906** (e.g., electric motors), a power bus electrically connecting the various power sources **904**, **906**, and a plurality of electrical devices **910** that may be powered by the engines **904** and/or electric machines **906**. The power system **900** includes a power distribution system **912** that distributes power **914** through power lines or cables **916**. The electric machines **906** may be configured as the aircraft electric motors shown and described herein and/or incorporate features as described herein.

(67) Advantageously, embodiments of the present disclosure provide for improved electric machines for aircraft and aviation applications. The aircraft electric machines of the present disclosure may provide for electric machines having reduced machine weight, increased efficiency, increased power density, and increased manufacturability. Advantageously, embodiments of the present disclosure are directed to fan assemblies integrated into the rotor assemblies of the electric machines to reduce Eddy current losses and to improve cooling of components of the electric machines (e.g., stator components, rotor components). In accordance with embodiments of the present disclosure, a fan assembly may be integrated in the rotor assembly. As such, rotation of the rotor assembly will cause rotation of fan blades of the fan assemblies, which in turn will draw or induce a cooling flow through the rotor assembly and around the stator assembly installed within the rotor assembly. In some embodiments, the fan assembly may be arranged at an outer diameter end of the rotor assembly (e.g., at an outer diameter end of an end portion of the rotor assembly). In some embodiments, in combination with an outer diameter fan assembly or as an alternative, a fan assembly may be arranged at an inner diameter end of the rotor assembly. Such fan assemblies may improve airflow through the electric machine and may enhance extraction of heat loads generated due to electromagnetic losses from the rotor and stator assemblies.

(68) In accordance with some embodiments of the present disclosure, the fan assemblies are arranged as structure features within or as part of an end portion of a rotor assembly, and thus may provide structural support and load transfer between various rotor components (e.g., between outer diameter, inner diameter, and end portions of the rotor assembly). The fan blades of the fan assemblies of the present disclosure may be shaped to improve efficiency and shape-optimized for Additive manufacturing, or the like. The fan assemblies, and the fan blades thereof, may ensure a level pressure drop to reduce uneven flow across the fan assemblies and to ensure even cooling flow through a stator cavity and thus provide cooling to surfaces/components of both a stator assembly and a rotor assembly of the electric machine. The operation of the fan assemblies and the cooling flow through the stator cavity may extract heat from rotor assembly surfaces and stator assembly surfaces and minimize hot spot temperatures. Accordingly, advantageously, embodiments of the present disclosure may enable increased power density and reduced magnet operating temperatures for electric machines.

(69) The terms “about” and “substantially” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” or “substantially” can include a range of $\pm 8\%$ or 5% , or 2% of a given value. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one

or more other features, integers, steps, operations, element components, and/or groups thereof. (70) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure is not limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. An aircraft electric motor comprising: a rotor assembly comprises an outer diameter portion, and inner diameter portion, and an end portion, arranged to define a stator cavity, and a rotor hub configured to connect to a shaft of the aircraft electric machine; a stator assembly arranged within the stator cavity; and at least one fan assembly defined in the end portion and configured to induce a cooling flow through the stator cavity, wherein the end portion includes an end portion support structure that houses end portion magnets and the rotor assembly further comprises: an outer diameter extension extending radially outward from the end portion support structure; an inner diameter extension extending radially inward from the end portion support structure; and the at least one fan assembly comprises: a first fan assembly arranged at an outer diameter end of the end portion and between the end portion support structure and the outer diameter extension; and a second fan assembly arranged at an inner diameter end of the end portion and between the end portion support structure and the inner diameter extension.
2. The aircraft electric motor of claim 1, wherein the at least one fan assembly is arranged at an outer diameter end of the end portion.
3. The aircraft electric motor of claim 1, wherein the at least one fan assembly is arranged at an inner diameter end of the end portion.
4. The aircraft electric motor of claim 1, wherein the at least one fan assembly comprises a first fan assembly arranged at an outer diameter end of the end portion and a second fan assembly arranged at an inner diameter end of the end portion.
5. The aircraft electric motor of claim 1, wherein the at least one fan assembly comprises a plurality of fan blades, wherein the plurality of fan blades provide structural support and load transfer within the rotor assembly between the inner diameter portion and the outer diameter portion through the end portion.
6. The aircraft electric motor of claim 1, wherein an inner diameter end of the end portion, the inner diameter portion, and the rotor hub are coupled together at a hub junction.
7. The aircraft electric motor of claim 6, further comprising a plurality of fasteners extending through and joining the end portion, the inner diameter portion, and the rotor hub together.
8. The aircraft electric motor of claim 1, wherein the outer diameter portion is coupled to the end portion by a plurality of fasteners.
9. The aircraft electric motor of claim 1, wherein: the outer diameter portion comprises a set of outer diameter magnets, the inner diameter portion comprises a set of inner diameter magnets, and the end portion comprises a set of end portion magnets.
10. The aircraft electric motor of claim 9, wherein the outer diameter magnets, the inner diameter magnets, and the end portion magnets are arranged in a substantially U-shaped configuration defining the stator cavity.
11. The aircraft electric motor of claim 1, wherein the outer diameter portion comprises a flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly, wherein the flow control feature is configured to direct a cooling flow into

the stator cavity.

12. The aircraft electric motor of claim 1, wherein the inner diameter portion comprises a flow control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly, wherein the flow control feature is configured to direct a cooling flow into the stator cavity.

13. The aircraft electric motor of claim 1, wherein the outer diameter portion comprises a first flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly and the inner diameter portion comprises a second flow control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly, wherein each of the first flow control feature and the second flow control features is configured to direct a cooling flow into the stator cavity.

14. The aircraft electric motor of claim 1, wherein the stator assembly comprises a plurality of windings arranged within the stator cavity.

15. An aircraft comprising: at least one aircraft electric machine; at least one electrical device; and a power distribution system configured to distribute power from the at least one electric machine to the at least one electrical device, wherein the at least one aircraft electric machine comprises: a rotor assembly comprises an outer diameter portion, and inner diameter portion, and an end portion, arranged to define a stator cavity, and a rotor hub configured to connect to a shaft of the aircraft electric machine; a stator assembly arranged within the stator cavity; and at least one fan assembly defined in the end portion and configured to induce a cooling flow through the stator cavity, wherein the end portion end portion includes an end portion support structure that houses end portion magnets and the rotor assembly further comprises: an outer diameter extension extending radially outward from the end portion support structure; an inner diameter extension extending radially inward from the end portion support structure; and the at least one fan assembly comprises: a first fan assembly arranged at an outer diameter end of the end portion and between the end portion support structure and the outer diameter extension; and a second fan assembly arranged at an inner diameter end of the end portion and between the end portion support structure and the inner diameter extension.

16. The aircraft of claim 15, wherein the stator assembly comprises a plurality of windings arranged within the stator cavity.

17. The aircraft of claim 15, wherein the at least one fan assembly comprises at least one of a fan assembly arranged at an outer diameter end of the end portion and a fan assembly arranged at an inner diameter end of the end portion.

18. The aircraft of claim 15, wherein the at least one fan assembly comprises a plurality of fan blades, wherein the plurality of fan blades provide structural support and load transfer within the rotor assembly.

19. The aircraft of claim 15, wherein: the outer diameter portion comprises a set of outer diameter magnets, the inner diameter portion comprises a set of inner diameter magnets, and the end portion comprises a set of end portion magnets.

20. The aircraft of claim 15, wherein, at least one of: the outer diameter portion comprises a first flow control feature extending from an axial end of the outer diameter portion opposite the end portion of the rotor assembly and configured to direct a cooling flow into the stator cavity along an outer diameter surface of the stator assembly, and the inner diameter portion comprises a second flow control feature extending from an axial end of the inner diameter portion opposite the end portion of the rotor assembly and configured to direct a cooling flow into the stator cavity along an inner diameter surface of the stator assembly.
